$$\dfrac{\vdash \Gamma, I \quad \vdash I^{\perp}, \Delta}{\vdash \Gamma, \Delta}\,(\mathsf{Cut})$$

$$\longrightarrow_{\mathsf{cut}} \quad \dfrac{\dfrac{\dfrac{\vdash F, \Gamma, I_1}{\vdash F, \Gamma, I}\,(\oplus^1) \qquad \vdash I^{\perp}, \Delta}{\vdash F, \Gamma, \Delta}\,(\mathsf{Cut}) \qquad \dfrac{\dfrac{\vdash G, \Gamma, I_2}{\vdash G, \Gamma, I}\,(\oplus^2) \qquad \vdash I^{\perp}, \Delta}{\vdash G, \Gamma, \Delta}\,(\mathsf{Cut})}{\vdash F \& G, \Gamma, \Delta}\,(\&)$$

$$\longrightarrow_{\mathsf{cut}} \quad \dfrac{\dfrac{\vdash F, \Gamma, I_1 \quad \vdash I_1^{\perp}, \Delta}{\vdash F, \Gamma, \Delta}\,(\mathsf{Cut}) \qquad \dfrac{\dfrac{\vdash G, \Gamma, I_2}{\vdash G, \Gamma, I}\,(\oplus^2) \qquad \vdash I^{\perp}, \Delta}{\vdash G, \Gamma, \Delta}\,(\mathsf{Cut})}{\vdash F \& G, \Gamma, \Delta}\,(\&)$$

$$\longrightarrow_{\mathsf{cut}} \quad \dfrac{\dfrac{\vdash F, \Gamma, I_1 \quad \vdash I_1^{\perp}, \Delta}{\vdash F, \Gamma, \Delta}\,(\mathsf{Cut}) \qquad \dfrac{\vdash G, \Gamma, I_2 \quad \vdash I_2^{\perp}, \Delta}{\vdash G, \Gamma, \Delta}\,(\mathsf{Cut})}{\vdash F \& G, \Gamma, \Delta}\,(\&) \quad = \pi$$